

Semester Project: Applying Network Economics to Real-World Settings

1. Project Pedagogy/Aim

In this project we aim to create an environment similar to one you would encounter in a workplace. It is designed around the idea that the project team (you) are in a workplace with a manager(s) (Yves and Arthur). Your job is to conduct an analysis that informs the manager about some aspect of the real world using the tools of network economics. You and your team are free to use any tools (AI included) to assist you in your analysis, as you could in a regular workplace. However, you must be able to explain and communicate whatever you come up with in a compelling manner to your manager(s). Importantly, you must be able to explain it in a way that a manager who has not taken ECC3841 (it is very unlikely that your manager will have taken our course) can understand what you have done and why it is interesting.

The project is quite open ended in the sense that you are free to pursue a topic or area that you are interested in. We encourage students to think innovatively and pursue something that they find valuable.

2. Project Overview & Core Question

In this course, we study formal tools to analyze how connections, structures, and strategic positions shape economic behavior. The objective of this semester-long project is to move from textbook theory to empirical reality: you will select a real-world setting, measure and map its underlying network structure using tools from class, and uncover a meaningful insight about how the network actively shapes human/firm behavior or market outcomes.

The Fundamental Question: *"In your chosen setting, how does network structure determine what individual agents choose to do, or allow us to understand the outcomes that occur?"*

Working in teams of 3-4, your goal is tell us something interesting about the world. You should apply specific methodological tools developed in class to show that network structure is a primary driver or predictor of economic outcomes—such as pricing, equilibrium selection, technology adoption, systemic risk, or information diffusion. You should then communicate it in a way that would be understood by someone who has not necessarily taken this course such as you may have to do in a job where you have a manager who may be familiar with the particular setting that you have chosen but not necessarily the techniques. Hence, an important part of the project also involves being able to communicate/explain what you are doing and why

3. Deliverables

As in any workplace you will be expected to regularly keep your manager up to date with what you are doing and provide a final report on your analysis.

Week	Deliverable	Milestone Focus
6 - 7	Presentation 1 in-class (10 minutes)	Define the area and focus of your project. What is it that you are trying to understand and how to you plan going about it?
8 - 9	Presentation 2 in-class (10 minutes)	Progress report and preliminary analysis. What have you done so far and what have you found?
10, 11, 12	Presentation 3 in-class (15 minutes)	Select specific economic models from class (e.g., threshold contagion, multi-sided pricing) to analyze agent incentives.
October 23rd	Project Report (10 pages)	Submit Final Written Report ()

4. Grading

The Project is worth 40% of the overall class grade. Each group must complete 3 presentations and for each presentation must submit their slides for grading. Each group must also submit a final written report. The breakdown of the grading for the project (in terms of the overall grade) is as follows

- **Presentation (15%)**
- **Slides (5%)**
- **Final Written Report (20%)**